

Voyage Planner

Plan a trip across the solar system: routes, launch windows, and travel time.

The Idea

Pick an origin and a destination (Earth to Mars, Earth to Jupiter’s moon Europa). Voyage Planner computes a transfer orbit, shows the travel time, the fuel needed (delta-v), and the next good launch windows, then animates the trajectory. Grok Voice acts as the ship’s AI explaining the plan, and Grok Imagine creates a “postcard” for each leg of the trip. You can also replay real missions such as Voyager and New Horizons.

Track Fit

Built with Cursor	Orbital mechanics code, animation, and UI built in Cursor.
Grok Voice / Imagine	Imagine postcards, Voice ship AI.
Real space data	JPL Horizons planet positions; real mission trajectories.
Navigation theme	Interplanetary navigation.
Also eligible	Big Red, Software, Design.

Features

MVP

- Choose origin and destination planets.
- Compute a Hohmann (minimum-energy) transfer: delta-v and travel time.
- Find the next launch window from the planets’ relative positions.
- 2D animated top-down view of the orbits and the transfer path.
- Grok Voice explains the plan in plain language.

Stretch

- Porkchop plot (launch date vs. arrival date vs. fuel cost).
- Gravity-assist routes (e.g. Earth–Venus–Jupiter).
- Replay Voyager 2 or New Horizons from real trajectory data.
- Grok Imagine postcards for each stop.

Tech Stack and Data

- Math: Python (poliastro-style calculations or hand-written) or JS.
- Data: JPL Horizons API for planet and spacecraft positions.
- Frontend: React with canvas/SVG, or three.js for 3D.
- AI: Grok Imagine and Grok Voice APIs.

Limitations and Risks

- **Hard math.** Anything beyond a simple Hohmann transfer (porkchop plots, gravity assists) needs a Lambert solver and careful orbital mechanics. It is easy to get wrong under time pressure.
Mitigation: Ship the Hohmann version first. Use an existing, tested library for anything harder.
- **Simplified model.** Hohmann transfers assume circular, flat (coplanar) orbits. Real missions differ, sometimes a lot.
Mitigation: Label results as approximations; compare to a real mission's numbers as a sanity check.
- **Library availability.** `poliastro`, a popular Python orbital library, is archived and may not install cleanly on new Python versions.
Mitigation: Check installation Friday night; have a hand-written Hohmann fallback.
- **Weak live demo.** It's a planning tool on a screen; less interactive than the phone or globe ideas.
Mitigation: Make the animation and the ship-AI voice the centerpiece.
- **Originality.** NASA's Trajectory Browser and many educational tools exist.
Mitigation: Focus on the conversational, story-like experience rather than raw calculations.
- **Grok Imagine accuracy.** Postcards are artistic, not real images.
Mitigation: Label them clearly; pair with real NASA images where available.

Scorecard

Demo-able	3/5	Screen-only demo.
Buildable in 24h	3/5	MVP is fine; stretch goals are math-heavy.
Navigation theme	4/5	Clear fit.
Wow factor	3/5	Depends on visuals.
Technical risk (5 = riskiest)	3/5	Moderate (orbital math).

Verdict

Good if the team likes math and physics. Educational and on-theme, but the demo is less exciting than SkyPilot or Rover Route.